

Document Title: XR-500 Industrial Cooling Controller Installation Manual Product Name: XR-500 Industrial Cooling Controller Document Type: Installation Manual Revision Number: Rev 2.1 Document ID: MAN-XR500-INST-001 Date: October 14, 2024

1. **Product Overview** The XR-500 Industrial Cooling Controller is a precision-engineered control unit designed for closed-loop industrial refrigeration, chillers, and secondary loop fluid circulation systems. Integrating microprocessor-driven thermal regulation with advanced diagnostic capabilities, the unit regulates compressor staging, variable-frequency pump speeds, and electronic expansion valve (EEV) positioning to maintain stable system fluid discharge temperatures. The enclosure is constructed from powder-coated die-cast aluminum to ensure long-term resistance to environmental contaminants and industrial atmospheric exposure. The XR-500 is optimized for heavy-duty manufacturing environments, pharmaceutical processing plants, and food cold storage facilities.
2. **Technical Specifications**
 - Input Supply Voltage: 24V DC $\pm 10\%$ (Class 2 power supply required)
 - Maximum Power Consumption: 48 W
 - Primary Temperature Sensor Inputs: 4x PT1000 RTD (3-wire configuration)
 - Analog Control Outputs: 4x 0–10V DC / 4–20 mA configurable
 - Relay Output Rating: 250V AC, 5A resistive load
 - Maximum Allowable System Fluid Pressure: 2.8 MPa (28 bar)
 - Recommended Coolant Loop Flow Rate: 120 L/min (Nominal)
 - Flow Rate Range: 40 L/min to 180 L/min
 - Communication Interfaces: RS-485 (Modbus RTU), Dual Ethernet (BACnet/IP, Modbus TCP)
 - Operating Ambient Temperature: -10°C to 50°C
 - Ingress Protection Rating: IP65 (with sealed cable glands)
 - Weight: 3.4 kg
3. **Installation Requirements** Physical mounting must be performed on a flat, vertical surface capable of supporting at least 15 kg to account for dynamic vibration. Utilize four M6 stainless steel bolts through the external mounting flanges located on the rear chassis corners. Maintain a minimum clearance of 150 mm on all sides of the enclosure to permit adequate passive convective airflow and allow sufficient bending radius for wiring looms. All electrical connections must enter through the removable gland plate at the base of the unit. Ensure that liquid-tight IP65 cable glands are tightened to a torque specification of 3.5 N·m. High-voltage relay wiring must be routed through separate conduits from low-voltage signal and sensor conductors to prevent EMI coupling. Shielded twisted-pair cable (minimum 24 AWG, 120-ohm characteristic impedance) must be utilized for RS-485 serial communication, with the shield grounded at the controller chassis ground lug only.
4. **Operating Conditions** The XR-500 is engineered for continuous duty under industrial environmental parameters. Relative humidity must be maintained between 5% and 95%, non-condensing. The internal electronics are rated for operation up to 2000 meters above sea level without thermal de-rating; above 2000 meters, reduce maximum

ambient operating temperature by 0.5°C per 100 meters. Exposure to direct sunlight should be minimized to prevent radiant solar heating of the enclosure. The system fluid loop must utilize a glycol-water mix (up to 50% propylene glycol by volume) or clean treated industrial water. The maximum allowable system fluid pressure must not exceed 2.8 MPa during steady-state or dynamic switching conditions. Operating below the minimum flow rate of 40 L/min will trigger low-flow safety interlocks and halt compressor staging.

5. **Safety Information** Only qualified industrial electrical technicians and certified HVAC/R specialists should perform installation, wiring, or commissioning of the XR-500. Hazardous AC voltages up to 250V AC may be present on the output relay terminal blocks even when the 24V DC logic supply is powered down, provided external load circuits remain energized. Always isolate all power sources using locked-out circuit disconnect switches before opening the terminal compartment cover. Electrostatic discharge (ESD) precautions must be taken when accessing internal board jumper configurations. Never apply supply voltage exceeding 30V DC to the main power terminals, as permanent board trace damage and capacitor rupture may occur.
6. **Maintenance / Service Procedures** Prior to initial power-up, conduct a thorough inspection of all mechanical mountings, electrical terminal screws, and gland seal integrity. Verify that terminal block screw connections are torqued to 0.5 N·m. Visually inspect the internal enclosure for condensation, metal shavings, or loose wire strands. Calibrate PT1000 temperature sensor inputs annually using an accurate reference standard. Check cable gland tightness and enclosure gasket compression every six months in high-vibration installations.
7. **Troubleshooting**
 - Symptom: Display dark, status LED inactive. Possible Cause: Loss of 24V DC input supply voltage or reversed polarity. Action: Measure DC voltage across terminals +V and -V. Ensure voltage is within 21.6V DC to 26.4V DC range. Correct polarity if reversed.
 - Symptom: Error Code E-102 (System Overpressure Warning). Possible Cause: Fluid loop pressure exceeding safe operating threshold. Action: Inspect system pressure gauge. Verify fluid pressure is below the 2.8 MPa limit. Check downstream motorized valves and pump operations.
 - Symptom: Error Code E-204 (Low Flow Alarm). Possible Cause: Coolant loop flow rate below 40 L/min threshold. Action: Check line strainers for debris, inspect pump impeller, and balance flow control valves to restore nominal 120 L/min circulation.
 - Symptom: RS-485 Communication Error. Possible Cause: Incorrect termination resistor configuration or ground loop. Action: Ensure 120-ohm end-of-line resistor is active only at the terminal node of the bus network. Verify single-point shield grounding.
8. **Additional Notes** Firmware updates are conducted via the primary Ethernet port utilizing the XR-500 Configuration Utility tool (v3.0 or higher). Always back up custom parameter tables before uploading new firmware binaries. The internal lithium coin cell battery (CR2032) preserves the real-time clock during power outages and has an estimated service life of 5 years under continuous operation. Dispose of depleted batteries in accordance with local environmental regulations.

